DDL: 5 月 12 日

8.1a

6.1 Define the following terms: phase, constituent, component, and degree of freedom.

8.4b

6.1(b) At 90°C, the vapour pressure of 1,2-dimethylbenzene is 20 kPa and that of 1,3-dimethylbenzene is 18 kPa. What is the composition of a liquid mixture that boils at 90°C when the pressure is 19 kPa? What is the composition of the vapour produced?

8.6a

6.3(a) It is found that the boiling point of a binary solution of A and B with $x_A = 0.6589$ is 88°C. At this temperature the vapour pressures of pure A and B are 127.6 kPa and 50.60 kPa, respectively. (a) Is this solution ideal? (b) What is the initial composition of the vapour above the solution?

8.10a

6.7(a) Blue $CuSO_4 \cdot 5H_2O$ crystals release their water of hydration when heated. How many phases and components are present in an otherwise empty heated container?

8.10b

6.7(b) Ammonium chloride, NH_4Cl, decomposes when it is heated. (a) How many components and phases are present when the salt is heated in an otherwise empty container? (b) Now suppose that additional ammonia is also present. How many components and phases are present?